Mobile Weather Monitoring System

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Abstract

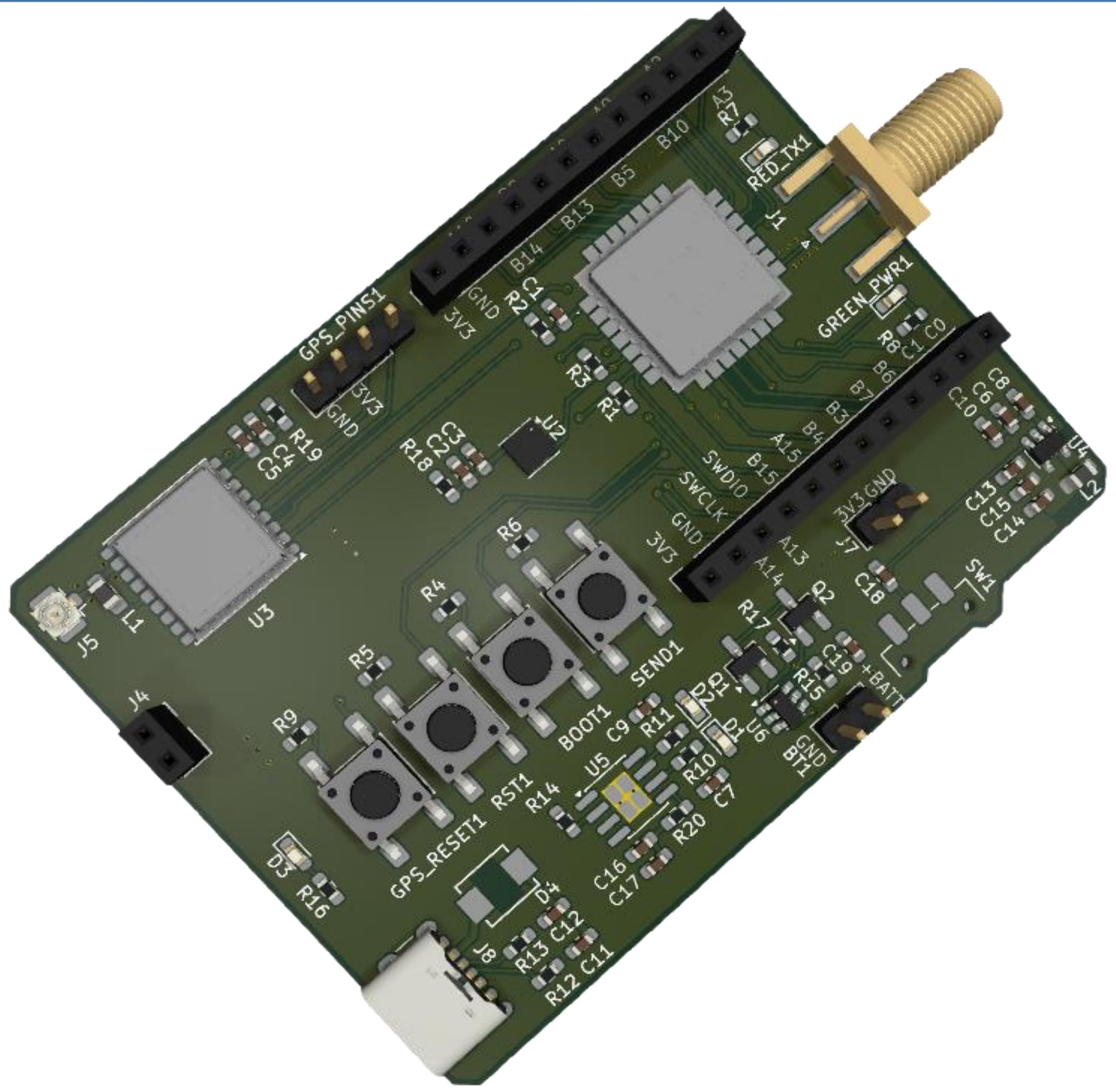
The Mobile Weather Monitoring System (MWMS) is a portable sensor for atmospheric data including temperature, humidity, pressure, and air quality index. This system correlates this data with its GPS coordinates and sends this data to the base station using LoRa communication. The base station then plots this data onto a map where each point is the location where the data was measured. This is designed for off-grid use, as we use LoRa peer-to-peer connection and offline maps to plot the data.

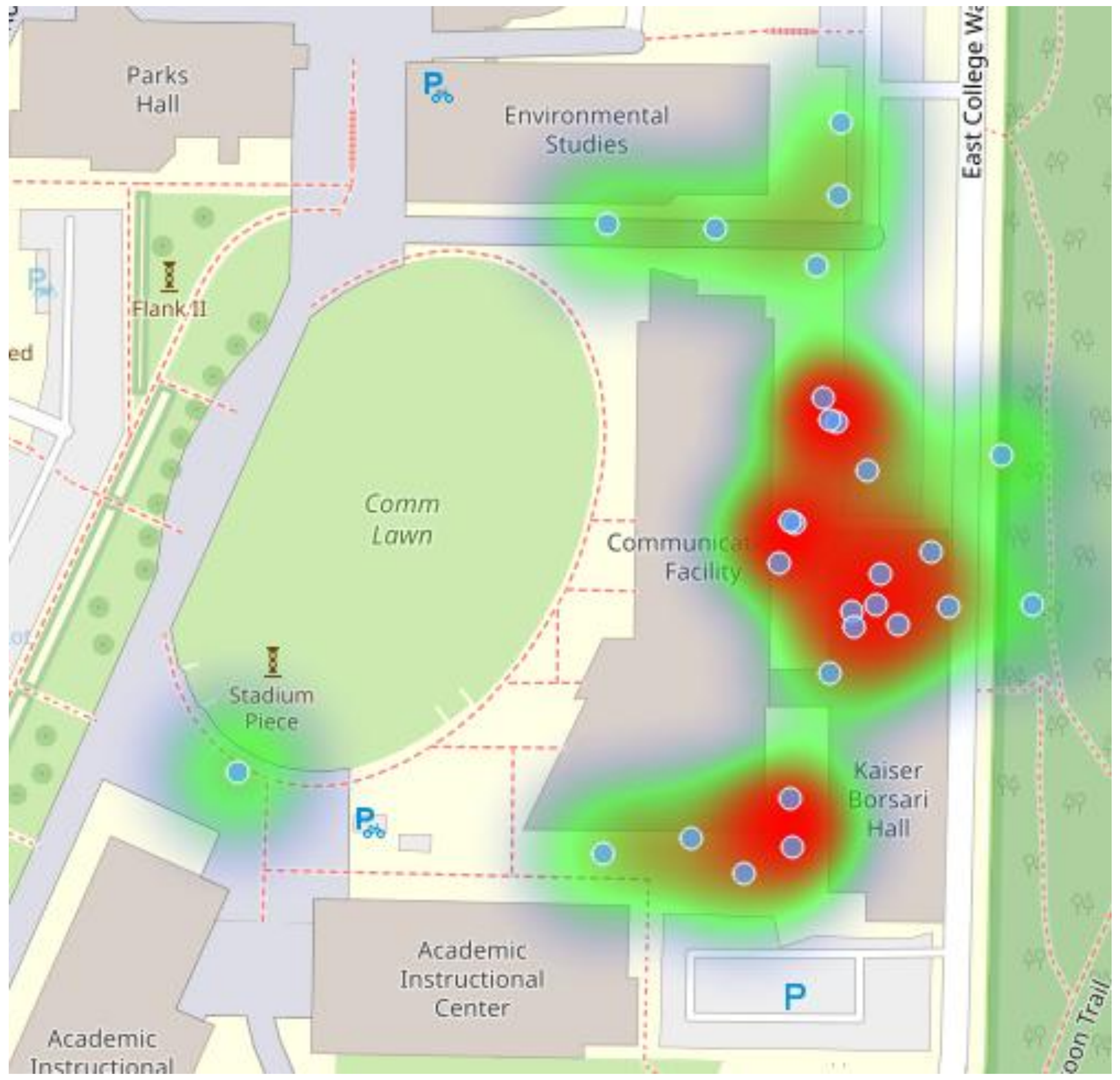
Methods and Materials

Hardware: u-blox MAX-M10S GPS Module, Bosch BME680 4-in-1 sensor, WIO-E5 STM32WLE5 wireless module, custom power supply.

Software: STM32CubeIDE, STM32CubeProgrammer, GitHub, VSCode, PuTTY, FreeRTOS, KiCAD, GIMP, SDRConnect.

Tools: STLINK-V2, 1850 battery, 915MHz antenna, 1.575GHz antenna, SDRPlay RSPdx-M2, soldering iron, reflow oven, soldering stencil, microscope, oscilloscope, multimeter, power supply, computer.





Results

Communication between the base station and the MWMS has been tested up to 4 km (2.49 miles) from Sehome Arboretum View Tower to Little Squalicum Pier. These tests also reported us that the RSSI is within LoRa detection meaning that our system can communicate farther than what we required. The sensor returns the right values for the atmospheric data when tested in a controlled room. GPS acquisition works extremely well and is very accurate outside.

Future Direction

- Adding battery backup for GPS configuration
- Swapping for an updated weather sensor
- Adding a laser particulate matter sensor
- Adding solar panel
- Continued GPS module development
- Improved battery connection and fuel gauge

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